

```
In [1]: #Importing lib
import numpy as np
import pandas as pd
```

```
In [2]: dataset = pd.read_csv("twitter.csv")
```

```
In [3]: dataset
```

Out[3]:

	Unnamed: 0	count	hate_speech	offensive_language	neither	class	tweet
0	0	3	0	0	3	2	!!! RT @mayasolovely: As a woman you shouldn't...
1	1	3	0	3	0	1	!!!! RT @mleew17: boy dats cold...tyga dwn ba...
2	2	3	0	3	0	1	!!!!!! RT @UrKindOfBrand Dawg!!!! RT @80sbaby...
3	3	3	0	2	1	1	!!!!!!! RT @C_G_Anderson: @viva_based she lo...
4	4	6	0	6	0	1	!!!!!!!!!!!! RT @ShenikaRoberts: The shit you...
...	...	...	...	...	...	...	...
24778	25291	3	0	2	1	1	you's a muthaf***in lie “@LifeAsKing: @2...

```
In [4]: dataset.isnull().sum()
```

```
Out[4]: Unnamed: 0      0
count      0
hate_speech 0
offensive_language 0
neither      0
class      0
tweet      0
dtype: int64
```

```
In [5]: dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24783 entries, 0 to 24782
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            24783 non-null  int64
1   count                 24783 non-null  int64
2   hate_speech           24783 non-null  int64
3   offensive_language    24783 non-null  int64
4   neither               24783 non-null  int64
5   class                 24783 non-null  int64
6   tweet                 24783 non-null  object
dtypes: int64(6), object(1)
memory usage: 1.3+ MB
```

In [6]: dataset.describe()

Out[6]:

	Unnamed: 0	count	hate_speech	offensive_language	neither	class
count	24783.000000	24783.000000	24783.000000	24783.000000	24783.000000	24783.000000
mean	12681.192027	3.243473	0.280515	2.413711	0.549247	1.110277
std	7299.553863	0.883060	0.631851	1.399459	1.113299	0.462089
min	0.000000	3.000000	0.000000	0.000000	0.000000	0.000000
25%	6372.500000	3.000000	0.000000	2.000000	0.000000	1.000000
50%	12703.000000	3.000000	0.000000	3.000000	0.000000	1.000000
75%	18995.500000	3.000000	0.000000	3.000000	0.000000	1.000000
max	25296.000000	9.000000	7.000000	9.000000	9.000000	2.000000

In [7]: dataset["labels"] = dataset["class"].map({0:"Hate Speech",
1:"Offensive Language",
2:"No Hate or offensive language"})

In [8]: dataset

Out[8]:

	Unnamed: 0	count	hate_speech	offensive_language	neither	class	tweet	labels
0	0	3	0	0	3	2	!!! RT @mayasolovely: As a woman you shouldn't...	No Hate or offensive language
1	1	3	0	3	0	1	!!!! RT @mleew17: boy dats cold...tyga dwn ba...	Offensive Language
2	2	3	0	3	0	1	!!!!!! RT @UrKindOfBrand Dawg!!!! RT @80sbaby...	Offensive Language
3	3	3	0	2	1	1	!!!!!! RT @C_G_Anderson: @viva_based she lo...	Offensive Language
4	4	6	0	6	0	1	!!!!!! RT @ShenikaRoberts: The shit you...	Offensive Language
...	...	...	...	...	...	...	...	...
24778	25291	3	0	2	1	1	you's a muthaf***in lie “@LifeAsKing: @2...	Offensive Language
24779	25292	3	0	1	2	2	you've gone and broke the wrong heart baby, an...	No Hate or offensive language
24780	25294	3	0	3	0	1	young buck wanna eat!!!... dat nigguh like I ain...	Offensive Language
24781	25295	6	0	6	0	1	youu got wild bitches tellin you lies	Offensive Language
24782	25296	3	0	0	3	2	~~Ruffled Ntac Eileen Dahlia - Beautiful col...	No Hate or offensive language

24783 rows × 8 columns

```
In [9]: data = dataset[["tweet", "labels"]]
```

```
In [10]: data
```

```
Out[10]:
```

	tweet	labels
0	!!! RT @mayasolovely: As a woman you shouldn't...	No Hate or offensive language
1	!!!! RT @mleew17: boy dats cold...tyga dwn ba...	Offensive Language
2	!!!!!! RT @UrKindOfBrand Dawg!!!! RT @80sbaby...	Offensive Language
3	!!!!!!! RT @C_G_Anderson: @viva_based she lo...	Offensive Language
4	!!!!!!!!!!!! RT @ShenikaRoberts: The shit you...	Offensive Language
...	...	...
24778	you's a muthaf***in lie “@LifeAsKing: @2...	Offensive Language
24779	you've gone and broke the wrong heart baby, an...	No Hate or offensive language
24780	young buck wanna eat!!.. dat nigguh like I ain...	Offensive Language
24781	youu got wild bitches tellin you lies	Offensive Language
24782	~~Ruffled Ntac Eileen Dahlia - Beautiful col...	No Hate or offensive language

24783 rows × 2 columns

```
In [11]: import re
import nltk
import string
```

```
In [12]: # Importing stop words
from nltk.corpus import stopwords
stopwords = set(stopwords.words("english"))
```

```
In [13]: # Importing stemming
stemmer = nltk.SnowballStemmer("english")
```

```
In [14]: def clean_data(text):
    text = str(text).lower()
    text = re.sub(r'https?://\S+|www\.\S+', '', text)
    text = re.sub(r'\[.*?\]', '', text)
    text = re.sub(r'<.*?>+', '', text)
    text = re.sub(r'[\{\}]{1,}'.format(re.escape(string.punctuation)), ' ', text)
    text = re.sub(r'\n', '', text)
    text = re.sub(r'\w*\d\w*', '', text)
    # Stop words removal
    words = [word for word in text.split(' ') if word not in stopwords]
    text = " ".join(words)
    # Stemming the text
    words = [stemmer.stem(word) for word in text.split(' ')]
    text = " ".join(words)
    return text
```

```
In [15]: data["tweet"] = data["tweet"].apply(clean_data)
```

C:\Users\ahmed\AppData\Local\Temp\ipykernel_28264\1832165696.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

```
data["tweet"] = data["tweet"].apply(clean_data)
```

```
In [16]: data
```

```
Out[16]:
```

	tweet	labels
0	rt mayasolov woman complain clean hous ...	No Hate or offensive language
1	rt boy dat cold tyga dwn bad cuffin...	Offensive Language
2	rt urkindofbrand dawg rt ever ...	Offensive Language
3	rt c g anderson viva base look li...	Offensive Language
4	rt shenikarobert shit hear mig...	Offensive Language
...	...	...
24778	muthaf lie lifeask pearl corey emanu...	Offensive Language
24779	gone broke wrong heart babi drove redneck crazi	No Hate or offensive language
24780	young buck wanna eat dat nigguh like aint ...	Offensive Language
24781	youu got wild bitch tellin lie	Offensive Language
24782	ruffl ntac eileen dahlia beauti color co...	No Hate or offensive language

24783 rows × 2 columns

```
In [17]: x = np.array(data["tweet"])  
y = np.array(data["labels"])
```

```
In [18]: x
```

```
Out[18]: array(['rt mayasolov woman complain clean hous amp man alway take trash ',  
'rt boy dat cold tyga dwn bad cuffin dat hoe place ',  
'rt urkindofbrand dawg rt ever fuck bitch start cri confus shi  
t',  
..., 'young buck wanna eat dat nigguh like aint fuckin dis',  
'youu got wild bitch tellin lie',  
'ruffl ntac eileen dahlia beauti color combin pink orang yellow amp w  
hite coll '],  
dtype=object)
```

```
In [19]: y
```

```
Out[19]: array(['No Hate or offensive language', 'Offensive Language',  
'Offensive Language', ..., 'Offensive Language',  
'Offensive Language', 'No Hate or offensive language'],  
dtype=object)
```

```
In [20]: from sklearn.feature_extraction.text import CountVectorizer  
from sklearn.model_selection import train_test_split
```

```
In [21]: cv = CountVectorizer()  
x = cv.fit_transform(x)
```

In [22]: x

Out[22]: <24783x25091 sparse matrix of type '<class 'numpy.int64'>'
with 198288 stored elements in Compressed Sparse Row format>

In [23]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size = 0.33, random_state

In [24]: x_train

Out[24]: <16604x25091 sparse matrix of type '<class 'numpy.int64'>'
with 132791 stored elements in Compressed Sparse Row format>

In [25]: *# Building out ML model*
from sklearn.tree import DecisionTreeClassifier

In [26]: dt = DecisionTreeClassifier()
dt.fit(x_train, y_train)

Out[26]: DecisionTreeClassifier()

In [27]: y_pred = dt.predict(x_test)

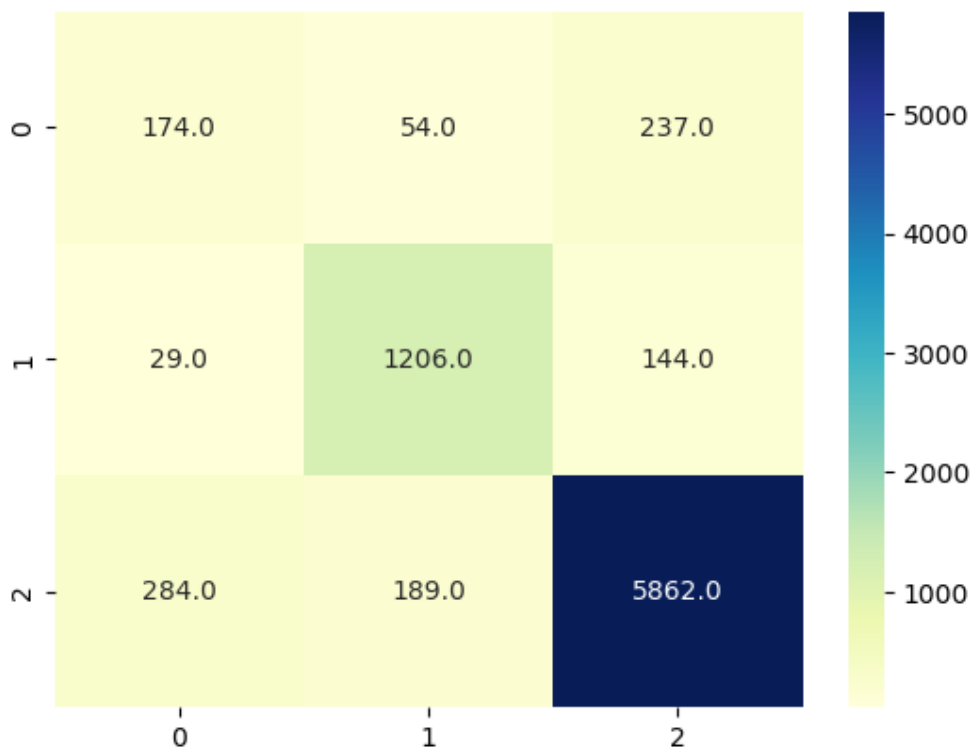
In [28]: *# Confusion Matrix & Accuracy*
from sklearn.metrics import confusion_matrix
cn = confusion_matrix(y_test, y_pred)
cn

Out[28]: array([[174, 54, 237],
[29, 1206, 144],
[284, 189, 5862]], dtype=int64)

In [29]: import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline

```
In [30]: sns.heatmap(cn, annot = True, fmt = ".1f", cmap = "YlGnBu")
```

```
Out[30]: <AxesSubplot:>
```



```
In [31]: from sklearn.metrics import accuracy_score  
accuracy_score(y_test, y_pred)
```

```
Out[31]: 0.8854383176427436
```

```
In [32]: sample = "Let's unite and kill all the people who are protesting against the government"  
sample = clean_data(sample)
```

```
In [33]: sample
```

```
Out[33]: 'let unit kill peopl protest govern'
```

```
In [34]: data1 = cv.transform([sample]).toarray()
```

```
In [35]: data1
```

```
Out[35]: array([[0, 0, 0, ..., 0, 0, 0]], dtype=int64)
```

```
In [36]: dt.predict(data1)
```

```
Out[36]: array(['No Hate or offensive language'], dtype=object)
```